

approximation for equity returns, an enormous literature formally rejects that equity returns are normally distributed.)

We summarize the probability density function by *moments*. Moments measure the shape of the probability density function, like where it is centered (*mean*), how disperse the returns can be (*variance* or *standard deviation*, which is the square root of the variance), how far the left-hand or right-hand tails extend (*skewness*), and how fat the tails are (*kurtosis*). The first four moments for the volatility strategy and S&P 500 are:

	Volatility Strategy	Equities
Mean	9.9%	9.7%
Standard Deviation	15.2%	15.1%
Skewness	−8.3	−0.6
Kurtosis	104.4	4.0

The volatility strategy and equities have approximately the same mean, around 10%. We see this in Figure 2.1 where the cumulated wealth of both strategies results in approximately the same ending value.⁵ While the standard deviation (or volatility—and I will use these two terms interchangeably) of the short volatility strategy and equities are similar at 15%, the long-left tail of the volatility strategy gives it very large negative skewness, of −8.3, and a humungous kurtosis of 104. The S&P 500 has small skewness of −0.6 and relatively small kurtosis of 4.0. In comparison, the normal distribution has a skewness of zero and kurtosis of 3.0. Hence, in Figure 2.2, Panel B, the normal distribution is a fairly close match for the distribution of equity returns.

2.2. RISK AVERSION

While the probability distribution of wealth shows us what returns are possible and how often, investors do not directly use these returns in making decisions. Instead, returns are used as inputs into a *utility function*, which defines how a return outcome is felt as a bad time or good time by the investor. Simply put, utilities convey the notion of “how you feel.” The utility function defines bad times for the investor.

Let us define the utility as a function of wealth, W , so that bad times are times when we are poor. (Technically we write $U(W)$, where utility, U , is a function of wealth, W .) The investor’s utility is not final wealth itself—the utility transforms wealth into an investor’s subjective value of wealth. Figure 2.3 plots three commonly used utility functions as a function of wealth: exponential, logarithmic, and constant relative risk aversion (CRRA), which is also called *power utility*.

⁵The correlation of the returns of the volatility strategy and equities is 0.51. This chapter only considers portfolio choice between single risky assets and a risk-free asset. Portfolio choice over multiple risky assets is considered in chapter 3.